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EXP - 12

TITLE - Stack Using Arrays

```
In [2]: def push(stack, item, max_size):
        if len(stack) >= max_size:
            print("Stack Overflow")

        else:

            stack.append(item)

            print("Pushed", item, "to stack")

def pop(stack):

    if not stack:
        print("Stack Underflow")

    else:

        item = stack.pop()

        print("Popped", item, "from stack")

        return item

def display(stack):

    if not stack:
        print("Stack is empty")

    else:

        print ("Stack elements are:")

        for item in reversed(stack):

            print(item)

max_size = int(input("Enter the maximum size of the stack: "))
stack = []

while True:
    print("=====")

    print("\n1. Push\n2. Pop\n3. Display\n4. Exit")
```

```

choice = int(input("\nEnter your choice: "))

if choice == 1:

    item = int(input("Enter the item to push: "))
    push(stack, item, max_size)

elif choice == 2:

    pop(stack)

elif choice == 3:

    display(stack)

elif choice == 4:
    break

else:

    print("Invalid choice, please try again...")

```

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 10 to stack

=====

1. Push
2. Pop
3. Display
4. Exit
Stack Overflow

=====

1. Push
2. Pop
3. Display
4. Exit
Popped 10 from stack

=====

1. Push
2. Pop
3. Display
4. Exit
Stack Underflow

=====

1. Push
2. Pop
3. Display
4. Exit

Stack is empty

=====

1. Push
2. Pop
3. Display
4. Exit

In []: